RETAIL SALES

AN END-TO-END DATA ANALYSIS PROJECT USING EXCEL, SQL, AND POWER BI

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Project Overview & Problem Statement

- **The Business Challenge**
  - Goal: To analyze the end-to-end operations of a retail company to improve sales strategies, optimize inventory, and monitor performance.
  - Data: Utilized a relational dataset with over 4,700 order items across 9 tables, covering sales, production, and inventory domains.
  - Key Questions: The project aimed to identify top-selling products, evaluate staff performance, track order fulfillment, and understand customer demographics.

Our Approach: The Data Pipeline

From Raw Data to Actionable Insights

1. Excel (Data Cleaning): Corrected inconsistencies, handled null values, and standardized formats to ensure data quality.
2. SQL (Database & Analysis): Loaded clean data into a relational MySQL database. Wrote complex queries with JOINs to answer key business questions and created 8 analytical Views.
3. Power BI (Visualization): Connected directly to the SQL Views to create an interactive dashboard for business users.

Dashboard Summary: Key Performance Indicators

- (Action: Take a screenshot of the three KPI cards from the top of your Power BI dashboard and place it here).
- Total Revenue: [e.g., $6.66M]
- Total Orders: [e.g., 1.4K]
- Total Customers: [e.g., 1.4K]

Key Takeaway: Provides an immediate, high-level overview of the business's scale and performance.

Insights

Revenue by Product Category

Mountain Bikes and Road Bikes Drive the Majority of Revenue

Finding: The top two categories, Mountain Bikes and Road Bikes, generate significantly more revenue than all other categories combined.

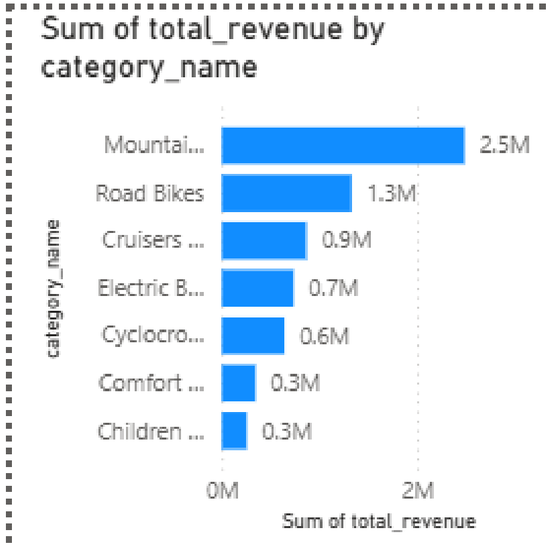
Implication: These are the most critical product lines for the business's financial health.

Sales Trends Over Time

Sales Show Strong Seasonality with Peaks in Q2 and Q4

Finding: The business experiences predictable peaks and troughs in order volume throughout the year. 2017 showed significant growth over 2016.

Implication: This allows for better planning of marketing campaigns and inventory stocking around peak seasons.



Insights

Customer Demographics

Customer Base is Heavily Concentrated in Three Key States

Finding: A vast majority of customers are located in New York (NY), California (CA), and Texas (TX).

Implication: Marketing and logistics efforts can be strategically focused on these high-value regions.

Top Performing Staff

Identifying Top Sales Performers

Finding: We can clearly identify the top-performing staff members based on the total sales revenue they have handled.

Implication: This data can be used for performance reviews, incentive programs, and identifying best practices to be shared across the team.

Sum of number_of_customers by state

Result Grid | Filter Rows: | Export:

first_name	last_name	store_name	total_sales_hand
Marcelene	Boyer	Baldwin Bikes	2405217.8490
Venita	Daniel	Baldwin Bikes	2295991.7173
Genna	Serrano	Santa Cruz Bikes	643627.9833
Mireya	Copeland	Santa Cruz Bikes	611863.6656
Kali	Vargas	Rowlett Bikes	375464.8000
Layla	Terrell	Rowlett Bikes	330449.2289

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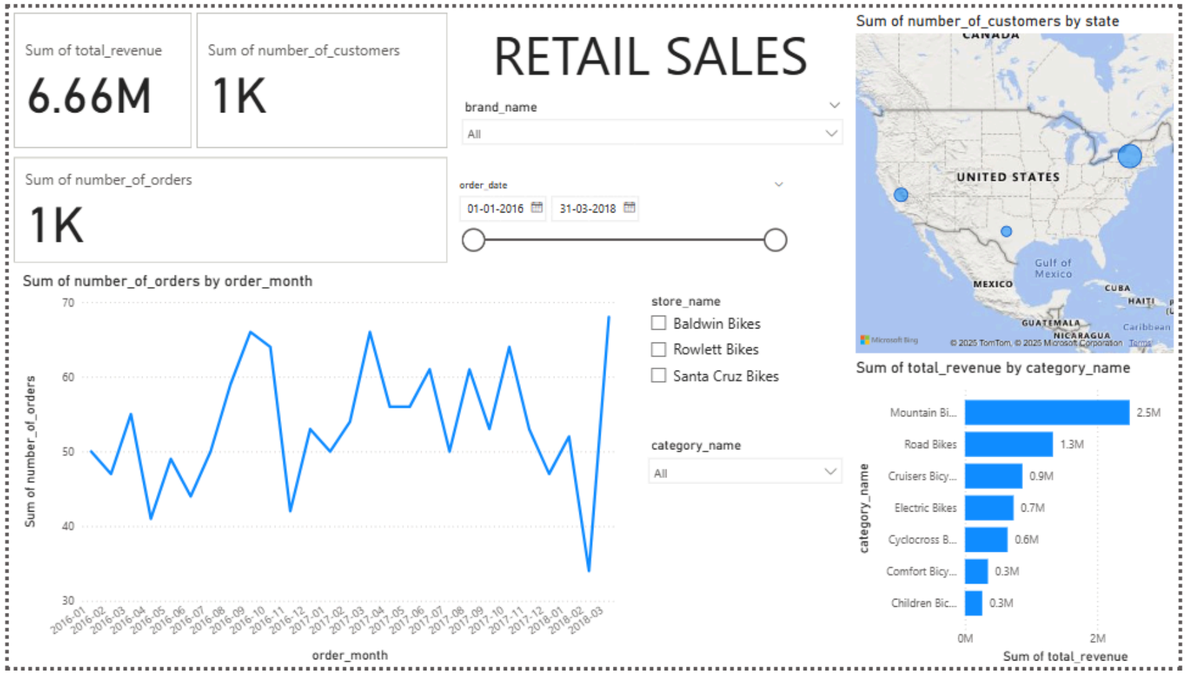
Recommendations for the Business

Actionable Recommendations from the Data

1. Marketing Strategy: Focus advertising spend and promotions on the **Mountain Bike** and **Road Bike** categories, especially in **New York** and **California**, to maximize ROI.

2. Inventory Management: Ensure high stock levels for top-selling brands (like Trek and Electra) in the top-performing stores (like Baldwin Bikes) to prevent stockouts, especially leading into peak sales months.

3.Operational Improvement: Analyze the delayed_shipments view to investigate the root causes of shipping delays and improve the customer experience.



Conclusion & Next Steps

Summary and Future Opportunities

- Conclusion:

By structuring our data and creating this interactive dashboard, we have provided clear, data-driven answers to critical business questions.

- Next Steps:

Customer Segmentation (Python): Apply machine learning (K-Means clustering) to segment customers into groups like "High-Value," "Loyal," and "At-Risk."

Predictive Analytics: Use historical sales data to build a forecasting model to predict future demand for key products.

THANK YOU

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